

History-Dependent Dynamics (HDD):

A Methodological Framework for Disentangling History Dependence, Recurrence, Self-Reference, and Self-Modeling in Dynamical Systems

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Abstract

Many physical, chemical, biological, neural, and artificial systems exhibit dynamics that depend on their previous states. Such phenomena are described using concepts including memory, adaptation, hysteresis, non-Markovianity, recurrence, and self-modeling. These concepts are related but are not equivalent, and evidence for one does not automatically establish the others. This paper introduces History-Dependent Dynamics (HDD), a falsifiable methodological framework for experimentally disentangling increasingly specific inferential claims about history-dependent organization. HDD begins with a basic empirical question: does information about the past improve out-of-sample prediction of future system behavior beyond the measured present state and current inputs? It then proposes progressively stronger tests addressing causal trajectory dependence, recurrent implementation, functional self-reference, and self-modeling. The framework treats these as diagnostic constructs rather than developmental stages, and classifies systems according to a profile $D = (d_1, d_2, d_3, d_4, d_5)$ relative to the quadruple (S, O, I, M) : the system, the observation model, the intervention set, and the mechanistic model class. HDD explicitly addresses the observation-model problem, in which apparent memory may arise because relevant latent variables are not included in the measured state. The framework is substrate-general and can be applied across biochemical networks, neural circuits, animal behavior, and artificial agents. We present a computational benchmark validating the discrimination of Construct I across four synthetic systems of known ground truth, with 100% accuracy across five random seeds and four noise levels. The complete framework is not yet fully empirically validated; HDD is presented as a falsifiable proposal whose empirical validity remains to be established.

Keywords: *history dependence; non-Markovian dynamics; memory; recurrence; self-reference; self-modeling; dynamical systems; causal inference; falsifiability*

1. Introduction

Many systems cannot be adequately described by considering their current measured state alone. A neuron may respond differently depending on its recent stimulation history. An animal's behavior may depend on previous interactions. A chemical reaction network may exhibit hysteresis and adaptation. These phenomena are commonly described using terms such as memory, adaptation, hysteresis, non-Markovianity, recurrence, and self-modeling. The problem is that these concepts are often discussed at different explanatory levels.

A system can exhibit history dependence without possessing an identifiable recurrent architecture. A recurrent system need not represent itself. A system can process information about itself without maintaining a model of its own dynamics. And even a demonstrable self-model does not, by itself, establish consciousness. The central methodological claim of this paper is therefore:

Evidence that a system depends on its history should not automatically be interpreted as evidence for more specific forms of dynamical organization. Stronger claims require stronger evidence.

HDD does not introduce a new estimator of temporal dependence. Its methodological contribution is the explicit separation of inferential claims that are frequently conflated: predictive history dependence, causal trajectory dependence, recurrent implementation, functional self-reference, and self-modeling. The framework specifies distinct empirical criteria and falsification conditions for each claim and proposes synthetic benchmarks designed to test whether these distinctions are identifiable.

Recent work by Leighton and Lynn (2025) developed information-theoretic decompositions of non-Markovian history dependence and demonstrated such dependencies in prolonged behavioral recordings. Leighton and Lynn (2026) further provide a tractable tunable model. The proposed contribution of HDD is not the discovery of history dependence itself—that is already an established research area—but the integration of these questions into a broader diagnostic architecture in which predictive history dependence is experimentally separated from causal, mechanistic, and self-referential claims. Where Leighton and Lynn provide tools for characterizing the structure of history-dependent prediction, HDD asks what additional evidence is needed to move from predictive to causal to mechanistic to self-referential claims.

Let X_t denote the measured state of a system at time t , P_t its current input or perturbation, and H_t a bounded representation of its preceding history. The first HDD question is whether:

$$P(X_{t+1} \mid X_t, P_t, H_t) \neq P(X_{t+1} \mid X_t, P_t)$$

If historical information improves out-of-sample prediction, the system exhibits history-dependent predictive structure. This result is deliberately weak. It does not establish a dedicated memory mechanism, recurrence, self-reference, self-modeling, agency, or consciousness.

1.1 The Observation-Model Problem

A system can be Markovian with respect to a sufficiently complete state representation while appearing non-Markovian when the observer measures only a coarse-grained subset of its variables. Suppose the true state is $Z_t = (X_t, L_t)$ where X_t is observed and L_t is an unmeasured latent variable. The complete system may satisfy $P(Z_{t+1} \mid Z_t)$ while the observed process satisfies $P(X_{t+1} \mid X_t, H_t)$ because history H_t provides indirect information about L_t . In such a case, history improves prediction without requiring a separate memory store.

This distinction is fundamental to HDD. A positive history-dependence result should always be described relative to the specified observation model. HDD does not treat statistical history dependence as proof of an ontologically distinct memory mechanism.

1.2 Scope and Organization

The paper is organized as follows. Section 2 introduces the five diagnostic constructs. Section 3 describes the evidential structure and null hypotheses. Section 4 specifies the statistical protocol. Section 5 presents the computational benchmark for Construct I. Section 6 discusses candidate empirical domains. Section 7 addresses what HDD does and does not claim. Section 8 specifies revision and falsification criteria. Section 9 concludes.

2. The Five Diagnostic Constructs

2.1 Overview

HDD proposes five constructs arranged as a diagnostic profile rather than a hierarchy. Each construct represents a distinct evidential commitment. Classification is relative to a quadruple (S, O, I, M): the system, the observation model, the intervention set, and the mechanistic model class. A given system may receive different HDD profiles under different experimental specifications, which is a feature rather than a limitation.

HDD classifies systems according to a profile $D = (d_1, d_2, d_3, d_4, d_5)$ where $d_i \in \{+, -, NI, UE, NT\}$:

Symbol	Meaning
+	Positive evidence for the construct
–	Evidence against the construct
NI	Not identifiable under available observations/interventions (established by formal identifiability analysis, not by failure to find evidence)
UE	Insufficient evidence or inadequate statistical power
NT	Not tested or not applicable

The ordering reflects increasing inferential specificity, not a developmental or computational sequence. A system may exhibit any combination of properties: history dependence without recurrence, recurrence without self-reference, self-reference without self-modeling.

	I	II	III	IV	V
Predictive evidence required	✓				
Causal intervention required		✓	✓	✓	✓
Mechanistic identification required			✓		
Self-specific representation required				✓	✓
Counterfactual own-dynamics model required					✓

Table 1. Evidential requirements for each construct.

2.2 Construct I — History-Dependent Predictive Structure

History-dependent predictive structure exists when information about preceding states improves prediction of future behavior beyond the measured present state and current inputs. Consider two models:

$$M_0: X_{t+1} = f(X_t, P_t)$$

$$M_1: X_{t+1} = f(X_t, P_t, H_t)$$

Define $\Delta L = L(M_0) - L(M_1)$ where L denotes out-of-sample loss. If ΔL is positive and exceeds prespecified statistical and practical significance criteria, the data support history-dependent predictive structure. Models must be compared under designs that control for differences in predictive capacity, including parameter count, regularization, and effective flexibility.

Construct I is observation-relative. It establishes predictive insufficiency of the measured state representation, not memory as an ontological mechanism. A negative classification applies only to the prespecified history family and temporal horizon.

Possible operationalizations include conditional mutual information, transfer entropy, Granger-style predictive comparisons, nonlinear forecasting, state-space models, and recurrent predictive models. These methods are not assumed interchangeable; different estimators capture different aspects of temporal dependence and should be reported as complementary.

A history dependence profile $\Delta L(\tau)$ computed across different history horizons τ shows at which temporal scales historical information is most predictive. An important control is the state reconstruction comparison: if a model M_S using a learned predictive state representation $S_t = f(H_t)$ eliminates the advantage of M_1 over M_0 , the dependence is reducible to state reconstruction under the specified model class.

2.3 Construct II — Causally Demonstrated Trajectory Dependence

Predictive dependence does not establish causation. A stronger test attempts to manipulate the prior trajectory while holding the present state and current perturbation as constant as experimentally possible. Consider two conditions with $X_t^A \approx X_t^B$, $H_t^A \neq H_t^B$, $P_t^A = P_t^B$, where H_t^A and H_t^B represent different experimentally induced prior trajectories. If $X_{t+1}^A \neq X_{t+1}^B$ systematically, the evidence for causal trajectory dependence is stronger.

This design estimates a conditional effect of manipulated trajectory on future dynamics given the measured present state. It does not estimate the total effect of history because matching on X_t may condition on a mediator of the historical effect. State matching must be formalized: a distance metric $d(X^A, X^B) < \epsilon$ must be specified with appropriate justification. State matching does not establish state identity; it establishes observational equivalence under the specified measurement model.

2.4 Construct III — Causally Identified Feedback Recurrence

History dependence does not imply recurrence. A system may depend on the past because a slow internal variable retains information without implementing feedback. Construct III concerns a specific mechanistic hypothesis: causally identified feedback recurrence at the level of system architecture, in which prior activity influences subsequent activity through an identifiable causal feedback pathway.

Evidence for recurrent implementation requires more than observing temporal dependence. A candidate recurrent pathway must be identified and experimentally perturbed. If disruption selectively reduces the history-dependent effect while controls demonstrate the intervention was effective, the evidence for recurrent implementation becomes stronger. A feedforward architecture with delay lines may produce the same input-output behavior as a recurrent network; therefore recurrent implementation is not generally identifiable from observational data alone.

2.5 Construct IV — Functional Self-Reference

Recurrence does not necessarily imply self-reference. A feedback system can recursively process information about an external environment without representing information about itself. HDD defines functional self-reference as:

Functional self-reference occurs when a representation whose content concerns the system's own state or dynamics is causally used to modify the system's dynamics in a manner specifically dependent on that content.

“Self” is an operational variable defined by the researcher, not a presumed metaphysical property. The system boundary and criterion for self-relevance must be specified before outcome analysis and independently of the effect being tested. A stronger experimental test compares self-

relevant information with non-self information matched as closely as possible along prespecified statistical and computational dimensions. If manipulation of self-relevant information produces effects not explained by generic information processing, evidence for functional self-reference is strengthened.

2.6 Construct V — Self-Modeling

Self-modeling is a stronger claim than self-reference. A candidate self-model is an internally maintained representation that allows the system to predict or regulate its own future dynamics—specifically its action-conditioned dynamics:

$$P(X_{t+1} \mid X_t, \text{do}(A_t = a), S_t^{\text{self}})$$

should be improved by the self-model relative to matched alternatives. HDD proposes three dimensions: V1 (self-prediction: the model predicts future states of the system itself); V2 (counterfactual self-prediction: the model predicts consequences of different actions on the system itself); and V3 (causal model use: manipulation of the model alters future planning or regulation in a corresponding manner). Evidence for self-modeling is strengthened when all three dimensions are satisfied.

A system may behave as if it predicts its own dynamics without maintaining a manipulable self-model. Construct V requires evidence that the representation can be causally manipulated and that this manipulation alters behavior consistently with the model’s predictions. This distinguishes a self-model from a conventional forward model or system-identification model.

3. Evidential Structure and Null Hypotheses

3.1 Evidential Summary

Construct	Empirical question	Positive interpretation	Main alternative explanation
I	Does history improve out-of-sample prediction?	Predictive history dependence	Omitted latent state
II	Does manipulated trajectory affect future conditional on measured state?	Residual causal trajectory dependence	State mismatch / latent mediation
III	Does feedback intervention selectively change the effect?	Causal recurrent implementation	Delay/feedforward equivalence
IV	Does self-relevant representation have specific causal role?	Functional self-reference	Generic information processing
V	Does representation predict/control own action-conditioned dynamics?	Self-modeling	Forward model / system identification

Table 2. Evidential structure for each construct.

3.2 Null Hypotheses

Construct	Null hypothesis
I	History does not improve out-of-sample prediction after controlling for state, input, and model capacity.
II	Manipulated trajectory has no effect conditional on the measured present state.
III	Disrupting the candidate feedback pathway does not selectively reduce the history-dependent effect.
IV	Self-relevant information has no effect beyond matched non-self information.
V	The candidate self-model does not improve counterfactual prediction or regulation of own dynamics beyond matched latent-state models.

Table 3. Null hypotheses for each construct.

3.3 The Inferential Prohibition

HDD contains a deliberate prohibition against inferential shortcuts. The chain:

History Dependence $\Rightarrow$ Recurrence $\Rightarrow$ Self-Reference $\Rightarrow$ Self-Modeling $\Rightarrow$ Consciousness

does not hold automatically. Each transition requires independent experimental evidence. A system may exhibit any combination of properties. Self-modeling (Construct V) entails a form of Construct IV by definition, but neither entails Construct III: a system may implement self-modeling through mechanisms other than the particular feedback recurrence tested under Construct III.

4. Statistical Protocol

4.1 Core Requirements

Out-of-sample evaluation (temporal cross-validation or held-out trajectories) is required for all constructs. Complexity-matched or appropriately penalized model comparisons are required for Construct I. Effect sizes, confidence intervals, and uncertainty measures must be reported alongside significance tests. Multiple comparison correction must be applied when testing across multiple history windows or representations.

4.2 The Analytical Pipeline Principle

The null must undergo the same relevant analytical procedure as the data:

Real: Raw $\rightarrow$ preprocessing $\rightarrow$ search $\rightarrow$ statistic

Null: Surrogate $\rightarrow$ same preprocessing $\rightarrow$ same search $\rightarrow$ statistic

A fully processed real-data statistic must not be compared against a null statistic generated through a materially different pipeline. When an analysis searches over multiple candidate parameters, the null distribution must reproduce the same search. If $T^* = \max_k T_k$ is selected from K candidate analyses, the appropriate comparison is T^*_{real} vs T^*_{null} , not $\max(T_{\text{real}})$ vs $T_{\text{null},k}$.

4.3 Discovery and Confirmation

When a parameter, lag, or model order is selected through empirical search, selection must occur on a discovery dataset. A temporally later confirmation dataset is then used to test the selected specification without further optimization. Information from the confirmation data must not influence model or parameter selection. A result that succeeds during discovery but fails confirmation must be reported as a failed confirmation.

4.4 Surrogate Design

A surrogate is valid only relative to the hypothesis it is intended to test. The surrogate should preserve the features that the null hypothesis says should remain while disrupting the relationship predicted by the alternative. A surrogate that destroys the wrong feature is not a valid null merely because it produces a convenient reference distribution.

4.5 Failure Modes

The following failure modes must be explicitly controlled: state aliasing (different real states appearing as the same X_t); common-cause history (history and future correlated through a third variable); selection bias; history representation bias; model capacity confound; intervention-induced state change; nonstationarity appearing as memory; distribution shift from train/test splits that do not respect temporal structure; and semantic labeling confound (where classification of a variable as “self-relevant” depends on researcher labeling rather than prespecified criteria).

5. Computational Benchmark: Construct I

5.1 Purpose

Before applying HDD to empirical systems, it must be validated on synthetic systems whose architectures are known. The benchmark reported here tests Construct I exclusively. Validation of Constructs II–V is designated as future work. The benchmark addresses the critical question: does the analysis produce correct classifications (+, −) relative to ground truth, and are improvements attributable to historical information rather than model capacity?

5.2 Synthetic Systems

Four ground-truth systems were implemented:

System	Dynamics	Expected HDD-I	Rationale
Markov	$X_{t+1} = 0.70 X_t + 0.40 P_t + \varepsilon$	−	Present state is sufficient; additional history provides no systematic gain
Hidden State	$z_{t+1} = 0.85 z_t + 0.35 P_t + \varepsilon$; $X_t = z_t + \delta$	+	History reveals latent variable not captured in observed X_t
Delay Line	$X_{t+1} = 0.55 X_t + 0.30 X_{t-3} + 0.30 P_t + \varepsilon$	+	Explicit lag-3 dependence not recoverable from X_t alone
Recurrent	$h_{t+1} = \tanh(Ah_t + BP_t + \varepsilon)$; $X_t = Ch_t + \delta$	+	History helps infer hidden recurrent state

Table 4. Ground-truth systems for Construct I benchmark.

The Hidden State system illustrates a critical methodological point: a positive Construct I classification for Hidden State does not imply a memory mechanism. History improves prediction because it provides information about a latent variable. This is precisely the observation-model problem that HDD is designed to make explicit.

5.3 Protocol

Pre-registered parameters: 120 training trajectories, 60 test trajectories, trajectory length 500, burn-in 100, history windows $\tau \in \{1, 2, 3, 5, 10\}$, noise levels $\sigma \in \{0.05, 0.10, 0.20, 0.30\}$, random seeds {42, 123, 456, 789, 2026}. Models: Ridge regression with α selected by trajectory-blocked 5-fold cross-validation on training data only. Inference: paired permutation test (5000 permutations) and paired bootstrap (2000 replications) with trajectory as the resampling unit. Classification criterion: HDD-I = “+” requires $CI_{n5}\%$ lower bound > 0 , relative improvement $\geq 1\%$, and $p < 0.05$.

Capacity control: a placebo history condition maintains the same number of historical features but destroys their temporal relationship to the target by row-permutation. Genuine historical information should produce larger improvements than the placebo. State reconstruction diagnostic: a model using PCA-compressed history ($n = 3$ components, fit on training data only) tests whether the advantage is reducible to low-dimensional state reconstruction.

5.4 Results

System	HDD-I	Expected	Relative improvement ($\tau=10$)	$CI_{n5}\%$	Permutation p
Markov	−	−	−0.012%	[−0.0056, +0.0033]	0.711
Hidden State	+	+	+9.60%	[+0.0241, +0.0276]	<0.001
Delay Line	+	+	+31.94%	[+0.0456, +0.0493]	<0.001
Recurrent	+	+	+6.75%	[+0.0142, +0.0166]	<0.001

Table 5. Construct I benchmark results at $\tau = 10$, noise = 0.10, seed = 42.

Accuracy against ground truth: 100% (4/4 systems correctly classified) at primary parameters. Robustness summary across all five seeds: Hidden State positive fraction 100%, Delay Line 100%, Recurrent 100%, Markov 0% (all seeds produced – or UE, never false +). Capacity control: for all non-Markov systems, real history improvement substantially exceeded placebo improvement (Hidden State: +9.6% real vs +0.03% placebo; Delay Line: +31.9% real vs –0.04% placebo; Recurrent: +6.7% real vs –0.06% placebo), confirming that improvements reflect genuine temporal information rather than model capacity. The state reconstruction diagnostic showed that PCA-compressed history did not recover the full predictive advantage for Hidden State or Recurrent, suggesting non-trivial temporal structure beyond low-dimensional state reconstruction.

5.5 Interpretation

The benchmark confirms that HDD Construct I can computationally discriminate systems with different ground-truth architectures, including the distinction between a genuinely Markovian system and systems whose history dependence arises from different mechanisms (latent state, explicit delay, recurrent dynamics). The positive classification for Hidden State does not imply memory as a mechanism—it illustrates exactly the observation-model problem: history improves prediction because it provides information about an unobserved variable.

This benchmark does not validate Constructs II–V. Those constructs require different experimental designs, including causal interventions and matched-state comparisons that are not available in the synthetic benchmark reported here.

6. Candidate Empirical Domains

The following cases illustrate how existing findings can be mapped onto HDD and identify where stronger tests would be required. These are candidate domains for HDD evaluation, not empirical validations of the complete framework.

6.1 Cellular and Biochemical Systems

History-dependent dynamics have been investigated in ion channel models, where observed history-dependent relaxation can arise from complex internal state structure not directly observable (Soudry & Meir, 2010). Microorganisms provide accessible history-dependent behavior: previous environmental exposure can alter subsequent cellular responses through metabolic, epigenetic, biochemical, and ecological mechanisms (Vermeersch et al., 2022). Chemical reaction networks have demonstrated dynamic switching in out-of-equilibrium molecular systems (Kriukov, Koyuncu & Wong, 2022). These systems provide candidate Construct I targets and illustrate that history dependence does not require biological complexity.

6.2 Neural Systems and Animal Behavior

Leighton and Lynn (2025) demonstrated history-dependent dependencies in prolonged fly behavioral recordings across timescales from fractions of a second to minutes. Their information-theoretic decomposition provides a methodological antecedent for HDD Construct I. HDD would not reinterpret these findings as evidence of self-modeling; instead, the next questions are: Can the relevant history dependence be causally manipulated (Construct II)? Does it depend on identifiable recurrent circuitry (Construct III)? Is any component specifically self-referential (Construct IV)?

6.3 Artificial Agents

Artificial systems provide useful test cases because aspects of their architecture can often be directly manipulated. Xiong et al. (2026) studied how memory addition and deletion affect LLM-agent behavior, finding that retrieved prior experiences influence subsequent outputs. These findings are relevant to Construct I motivation but insufficient to establish it under the matched-state criterion. If memory contents are included in the measured state, the system may be Markovian relative to that expanded representation. Recent work on egocentric visual self-modeling in robots (Hu et al., 2025) provides a closer antecedent for Construct V, demonstrating a predictive model of the robot’s own dynamics used for adaptation.

6.4 Open Quantum Systems

Non-Markovian dynamics are extensively studied in open quantum systems, where future evolution of a subsystem can depend on prior interactions with an environment (Breuer et al., 2016; Rivas et al., 2014). A quantum system can display strong memory effects without possessing anything resembling a self-model. HDD does not propose a new definition of quantum non-Markovianity; it identifies quantum systems as candidate Construct I domains while clarifying that positive Construct I classifications do not license inferences about higher constructs.

7. Scope Clarification

7.1 What HDD Claims

HDD claims that: history dependence should be distinguished from stronger dynamical properties; predictive and causal evidence should be separated; recurrence should be tested independently; self-reference should not be inferred from recurrence alone; self-modeling should be distinguished from generic latent-state estimation; these distinctions can be organized into a common experimental program; and existing empirical literatures provide multiple domains in which parts of the framework can be tested.

7.2 What HDD Does Not Claim

HDD does not claim that history dependence is a new discovery; that non-Markovianity is equivalent to memory in every ontological sense; that recurrence implies self-reference; that self-reference implies self-modeling; that self-modeling implies consciousness; that biological and artificial implementations are mechanistically identical; or that the framework has been empirically validated as a whole. HDD is not a theory of consciousness. Consciousness is treated only as a possible downstream application, requiring additional normative and empirical premises beyond what any HDD profile establishes.

7.3 Relationship to Prior Frameworks

HDD is not intended to replace existing theories of memory, dynamical systems, predictive processing, recurrence, self-modeling, or consciousness. Computational mechanics (Shalizi & Crutchfield, 2001) addresses the problem of finding minimal predictive state representations; its causal states provide a formal solution to part of what HDD identifies as the observation-model problem. Predictive state representations (Littman, Sutton & Singh, 2002) represent system state through predictions of future observations conditioned on actions, providing a methodological antecedent for Construct I state reconstruction. Self-model theories (Kwiatkowski et al., 2022; Hu et al., 2025; Limanowski & Blankenburg, 2013; Metzinger, 2003) have developed accounts of self-representation; HDD asks how self-modeling can be distinguished from latent-state estimation. The adversarial collaboration methodology of the Cogitate Consortium (2025) exemplifies the broader principle that theoretical constructs should generate experimentally discriminable predictions rather than being inferred retrospectively from compatible observations.

8. Revision and Falsification Criteria

8.1 Construct-Level Retirement

No construct is protected from negative evidence. A construct should be revised or removed if: it cannot be operationalized independently; it repeatedly fails appropriate controls; it cannot generate discriminating predictions; it collapses empirically into another construct without explanatory loss; or its apparent effects are consistently explained by simpler alternatives.

If computational validation shows that HDD cannot distinguish known architectures, the framework should be revised before being applied to empirical systems. If self-modeling cannot be distinguished from generic latent-state estimation under the proposed criteria, the definition of self-modeling should be revised or the construct removed. A failed extension should make the framework smaller, not more elaborate.

8.2 Framework-Level Falsification

The framework as a whole would be weakened if the five constructs systematically fail to provide independent discrimination across benchmark systems; if Construct I cannot be operationalized with adequate sensitivity and specificity across multiple independent domains; or if the evidential ladder proves systematically non-identifiable (NI) at levels II–V across all testable systems. In such cases, HDD would be restricted to its validated scope rather than abandoned entirely, consistent with the principle that the evidence determines how much of HDD remains necessary.

8.3 The Anti-Rescue Principle

The framework must not become impossible to falsify by continuously changing what counts as evidence. No new construct or metric may be introduced solely because an existing analysis failed to detect the predicted phenomenon. A new estimator is justified only when the previous estimator has a known limitation, the new estimator targets the same pre-specified construct or a clearly distinguished new construct, and the choice is made before confirmatory testing. Otherwise, metric proliferation becomes a form of theoretical escape.

9. Discussion and Conclusion

The central problem addressed by HDD is not the existence of memory—memory-like and history-dependent effects are already pervasive across scientific disciplines. The problem is inferential granularity. A researcher observes that the past matters and may describe the system as having memory. The system is recurrent, and the researcher may infer self-reference. The system processes information about itself, and the researcher may infer self-modeling. Each inference may be reasonable in a particular theoretical context, but none follows automatically from the previous observation.

HDD proposes that this inferential chain should be experimentally decomposed. This is particularly valuable because different scientific fields have developed different vocabularies for closely related phenomena. HDD’s contribution is to place these questions within a common methodological structure and ask what additional evidence is necessary to move from one explanatory level to the next. The framework does not claim that all instances of history dependence share a common mechanism, nor that the five constructs form an evolutionary or computational hierarchy.

The computational benchmark reported here demonstrates that Construct I can reliably discriminate systems with different ground-truth architectures, including the crucial distinction between a Markovian system and systems whose history dependence arises from latent states, explicit delays, or recurrent dynamics. This discrimination is robust across five random seeds and four noise levels. The Hidden State result is particularly important: it demonstrates that a positive Construct I classification does not establish memory as a mechanism, consistent with the framework’s foundational methodological claim.

Validating Constructs II–V requires experimental designs beyond synthetic benchmarks: causal trajectory manipulations, feedback perturbations, self-reference comparisons with matched-content controls, and counterfactual self-model tests. These designs are feasible in multiple candidate domains (neural systems, animal behavior, artificial agents, biochemical networks) and constitute the research program that HDD is designed to support.

The appropriate outcome is not that HDD survives at all costs. The appropriate outcome is that the evidence determines how much of HDD remains necessary. Whatever survives that process will not be the assumption of the framework—it will be what the data have allowed the framework to keep.

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